

About

Almost as a tradition, when someone is learning to code, the first program they are likely to write is the [Hello World program](#). So that's what you will be doing today! The real "assignment" here is more about installing software, setting up a programming environment, learning how to submit to gradescope, and the auto-grader, etc.

For information about how to submit on gradescope, check out section 7 below!

Learning Objectives

1. Install Python
2. Set up Integrated Development Environment (IDE)
3. Write your first program in python
4. Become familiar with Gradescope

Estimated Size

One file, with one line of code and three lines of comments.

Assignment

For this assignment, you must work alone. You are welcome to talk to people nearby, but everyone needs to make their own submission. We want everyone to have the necessary software installed and in working order, and to know what the Gradescope process looks like.

0.1. Install Python

Follow [the guide here](#) to install the Python program.

0.2. Set up VSCode

Follow [the guide here](#) to install and set up Visual Studio Code (VSCode).

1. Download this file named **hello.py**

Download it here :

<https://drive.google.com/file/d/1ZjilS5YZCdm1xW1ulQrhrXsKHdP2QKfb/view?usp=sharing>

- Click the download button  at the top of the screen
- Make a folder for this assignment somewhere of your choosing.
- Move the file hello.py into that folder
- Open that folder in VSCode (Open VSCode and select **File > Open Folder**, navigate to that folder you made and select it).
- There are a few issues with this file that will cause gradescope to fail, so let's see all of those and figure out how to fix them!

1.5. Try to run the program in VSCode

Before making the first submission to Gradescope, you should run the program on your computer to make sure its output is as expected. In VS Code you can click the menu: **Run -> Run without Debugging**. Alternatively, you should see a Run (triangle ▶) button on the top right of your program window. Click that also runs the program. A terminal window should pop up at the bottom and show the program output. (Note that if you didn't install the Python extension as in step 0.1 above, VS Code will prompt you to install the extension first).

You can also run the program in Terminal using the **python** command. To do so, select menu: **Terminal -> New Terminal** then type **python3 hello.py**

Either way, you should see the program prints out **Hello, World!** as its output. If so, you can proceed to the next step.

If you would like to increase the editor font size: go to menu **Code -> Settings -> Settings** and locate the **Editor Font Size** option and change it there.

2. Submit to Gradescope (1st attempt)

- Try to submit the file (as originally downloaded) to gradescope, more detailed instructions are included later in this document.
- You should see something like this

1.0) hello.py exists (0/5)

```
*** Looking for hello.py ***
```

```
Test Failed: 1 != 0 : Missing hello.py! ✖
Perhaps the file is not named correctly, it needs to match hello.py exactly
Or perhaps there is some critical error stopping the file from being read...
```

1.1) hello.py author (0/5)

```
*** Looking for hello.py's author info ***
```

```
Test Failed: 1 != 0 : Missing hello.py! ✖
Perhaps the file is not named correctly, it needs to match hello.py exactly
Or perhaps there is some critical error stopping the file from being read...
```

1.2) hello.py test (0/5)

```
*** Testing hello.py ***
```

```
Test Failed: 1 != 0 : Missing hello.py! ✖
Perhaps the file is not named correctly, it needs to match hello.py exactly
Or perhaps there is some critical error stopping the file from being read...
```

Lab 00: Code (INDIVIDUAL)

● Graded

Student

Unknown Student (removed from roster?)

Total Points

0 / 100 pts

Autograder Score

0.0 / 100.0

Failed Tests

1.0) hello.py exists (0/5)

1.1) hello.py author (0/5)

1.2) hello.py test (0/5)

- All the tests failed, and we got the same error for each one. We are told that the test failed while the autograder was **Looking for hello.py** and was told that the file is missing.
- This is because the file is not named correctly, if we scroll back to the top, and click on **Code** so we can see the file we submitted

Submitted Files

Results

Code

▼ hello.py.py

Download

- And we can see that the file is not named correctly, it's called **hello.py.py** but should be called **hello.py**
 - And yes, despite how annoying it may be at first, files need to be named **exactly** correctly, punctuation and capitalization included. The autograder can only find the file by looking for the name, and it can only find that name if it matches exactly.
- So rename the file using finder/file-explorer to remove the second **.py** and submit the file again

3. Submit to Gradescope (2nd attempt)

- Looking at gradescope, we should see that one test is passing now, we now have 5 out of 15 points! and the two failing tests have changed their error text, giving us another hint on how to proceed.
- You can see the screenshot below of how my gradescope looks after fixing the files name

```

1.0) hello.py exists (5/5)

*** Looking for hello.py ***

[✓] hello.py found

1.1) hello.py author (0/5)

*** Looking for hello.py's author info ***

Test Failed: False is not true : ✗ hello.py doesn't have enough lines to possibly have Author, Email, a

```

Lab 00: Code (INDIVIDUAL) ● Graded

Student

Unknown Student (removed from roster?)

Total Points

5 / 15 pts

Autograder Score

5.0 / 15.0

Failed Tests

1.1) hello.py author (0/5)

1.2) hello.py test (0/5)

Passed Tests

1.0) hello.py exists (5/5)

4. Add author information to **hello.py**

As the test 1.1 lets us know, we need to add an author comment to our file. We'll ask you to do this on all the code you submit, and for the first half of the class the grader will fail if you don't.

Open the file in VSCode, put your name, UMass email, Spire ID to help us link each submission with their email on Gradescope (sometimes non-UMass emails are used, and it's difficult to find the UMass one in that case)

That can be done with *comments*. Comments are lines that start with a **#** character. They are used as the programmer's notes and therefore are ignored by the computer.

So specifically, we want:

- **# Author:** to be the start of the first line of the file.

- **# Email:** to be the start of the second line of the file.
- **# Spire ID:** to be the start of the third line of the file.
- each of the above followed by the relevant information.
- Formatting needs to be exactly the same! **#** then a **space** then **Author** then a **colon** a **space** then your **name**
 - You will fail the gradescope test if it's not exact, double check and resubmit!!!

For example, the first three lines of a file may look like this:

```
# Author: John Snow
# Email: jsnow@umass.edu
# Spire ID: 12345678
```

Note: again, going forward this is expected in every file that you submit to Gradescope.

6. Submit to Gradescope (attempt 3)

1.0) hello.py exists (5/5)

```
*** Looking for hello.py ***
```

✓

hello.py found

1.1) hello.py author (5/5)

```
*** Looking for hello.py's author info ***
```

✓

Author information was included

✓

Email information was included

✓

Spire ID information was included

✓

hello.py author information found

1.2) hello.py test (0/5)

Lab 00: Code (INDIVIDUAL) ● Graded

Student

Unknown Student (removed from roster?)

Total Points

10 / 15 pts

Autograder Score

10.0 / 15.0

Failed Tests

1.2) hello.py test (0/5)

Passed Tests

1.0) hello.py exists (5/5)

1.1) hello.py author (5/5)

- We are passing the second test now!
- Now let's look at the text of the error text for test 1.2 and try to make sense of it.

1.2) hello.py test (0/5)

```
*** Testing hello.py ***

Test Failed: ' hello, world' != 'Hello, World!'
- hello, world
? ^^      ^
+ Hello, World!
? ^      ^      +
: Output does not seem to be "Hello, World!", instead it is " hello, world"
You can use the above text to identify what characters are different,
what needs to be added, and what needs to be removed to make the program produce correct output
```

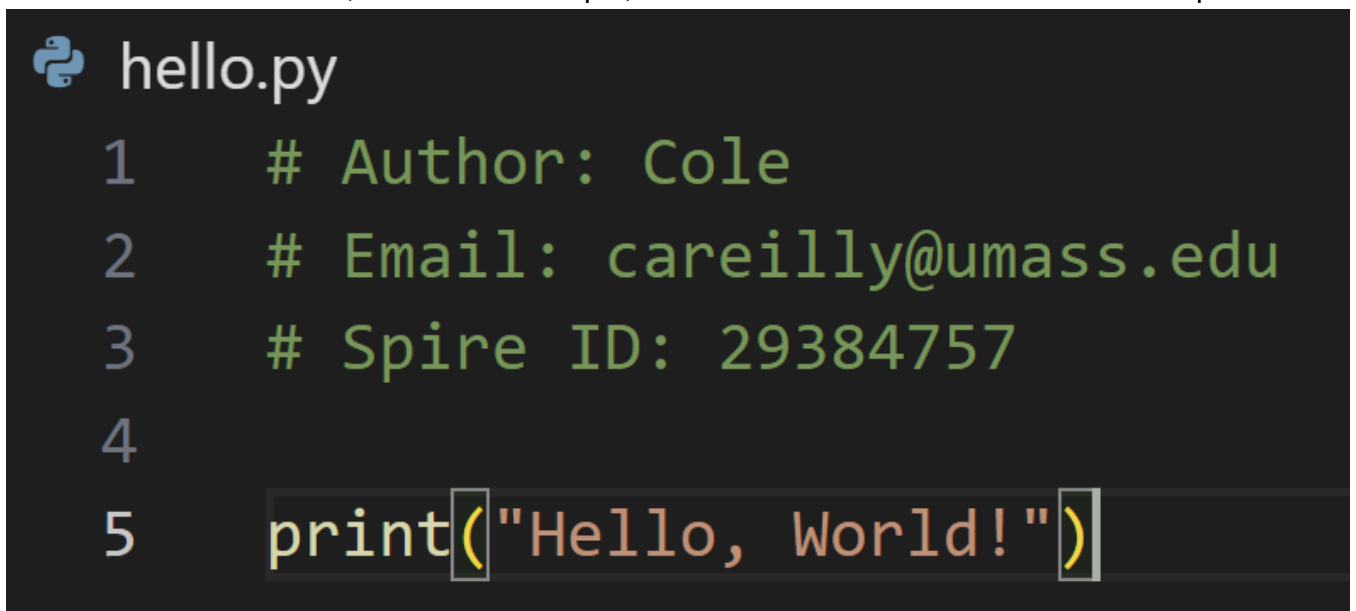
- This can be a little hard to read. Here is a general breakdown:
 - The first line (with a '-' in front of it) is the output our program produced.
 - The line below it (with a '?' in front of it) highlights the differences between the above line

and the below.

- a '^' means the character above it needs to change somehow
- a '-' means there is a character in the line that needs to be removed
- The third line (with a '+' in front of it) shows us the output our program **was expected to produce** (the desired output)
- The line below it (with a '?' in front of it) highlights the difference between the above line, and the first line.
-
- Now here is a breakdown explaining the specifics for this test

```
Test Failed: ' hello, world' != 'Hello, World!'
- hello, world
? ^^      ^
+ Hello, World!
? ^      ^      +
```

- Our output has an extra space character at the front (a ' ') it is marked with a '^' below it, showing us it needs to change
- Our output has a lowercase 'h', it is marked in our line and in the desired line with a '^' showing us we need to change our lowercase 'h' to an uppercase 'H'
- We have the same issue for the lowercase 'w', we need to change it to a 'W'
- In the expected output line, it ends with a '!' which is marked with a '+' below it, which means the character exists in this line, but not in our output, so we need to add a '!' to the end of our output



```
hello.py
1  # Author: Cole
2  # Email: careilly@umass.edu
3  # Spire ID: 29384757
4
5  print("Hello, World!")
```

This screenshot highlights all the changes we've made in the file to pass the autograder.

7. Running the program on your computer

Before making the final submission to Gradescope, you should run the program on your computer to make sure its output is as expected. You should avoid over-relying on the autograder, while our class

(110) is very forgiving, and allows infinite attempts, not every CS class will be like this. You should be able to solve all the problems and get a correct working program without needing the autograder.

Later in the semester there will be sections of projects with no autograder and you will have to test your program yourself. The final project will also have no autograder whatsoever.

8. Submit **hello.py** to Gradescope

Log in to [Gradescope](#), select the relevant assignment (**Lab 00: Code (Individual)**), and submit **hello.py** there.

If you run into any trouble submitting, gradescope has [a guide](#) on submitting [assignments like this](#).

Wait until the autograder finishes running and returns the result to you. You can resubmit as many times as you like before the deadline.

9. Fill out the Lab 0 Questions on Gradescope

Go back to the Gradescope assignments page, select the assignment **Lab 0: Questions**, and fill in your responses there. This is a necessary part of the lab and it can be edited as many times as you want before the deadline.

If you see a question you do not know how to answer, ask TAs/UCAs in your lab, or use piazza, or go to one of our office hours.

The assignment lets you submit your answers directly through gradescope, in the future make sure you submit with the names of your team members marked!

10. Additional Submission Instructions

Auto-Grader

We have set up Gradescope to use auto-grader to automate the grading process. Gradescope will look for files with specific names, and those names are **case-sensitive**. So if in the instructions we ask for a file named **hello.py**, your file must be named exactly that, with all letters in lower-case. For example, the following will NOT be recognized by the autograder: **Hello.py**, **hello.txt**, or **hello.py.txt**.

Please note that for this lab, **every student must submit individually**. No group submission is allowed for this lab. In the future when group submission is allowed, you will be given additional instructions.

Although this submission only involves one file, in the future, if your submission involves multiple program files, you need to submit them all at once, otherwise the auto-grader will return an error. You can do so in several ways:

- Select all files you want to submit, then drag and drop them in the submission area when prompted.
- Browse your files when prompted by Gradescope and select all files you want to submit.
- Alternatively, compress all the files you want to submit into a single zip file (the zip file name does not matter); then submit that zip file via drag-and-drop or browsing.
- You can also compress the folder that contains all the files you want to submit into a single zip file (the zip file name does not matter), and submit that zip file via drag-and-drop or browsing.

Unlimited attempts

You have unlimited attempts for our auto-graded assignments.

This means that you should submit early and often in order to be confident that the code you write will pass all of the autograder tests.

Academic Honesty

All work that is completed in this assignment is your own. You may talk to other students about the problems, and brainstorm about solutions. However, you may NOT share code in any way. What you submit **must be your own work**.

You may not use any code that is posted on the internet. If you are not sure it is in your best interest to contact the course staff. We will be using software that will compare your code to other students in the course as well as online resources. It is very easy for us to detect similar submissions and will result in a failure for the exercise or possibly a failure for the course. Please, do not do this. It is important to be academically honest and submit your work only. Please review the [UMass Academic Honesty Policy and Procedures](#) so you are aware of what this means.

Copying solutions, whether partial or whole, obtained from other students or elsewhere, is academic dishonesty. Do not share your code with your classmates, and do not use your classmates' code. If you are confused about what constitutes academic dishonesty you should re-read the course policies. We assume you have read the course policies in detail and by submitting this project you have provided your virtual signature in agreement with these policies.